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In [19]: #Session 2 Hoomework
# Name: Mohammed Akif
# Date: May 2025
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In [20]: # TASK 1: Lists (15 points)

# 1a. Create a List of 5 Bangladeshi cities
cities = ["Sylhet", "Rangamathi", "Bandorban", "Cumilla", "Dhaka"]

# 1b. Add "Rajshahi" to the end of the list
cities.append("Rajshahi")

# 1c. Remove the second city from the list
cities.pop(1)

# 1d. Print the list sorted alphabetically
print(sorted(cities))
```

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['Bandorban', 'Cumilla', 'Dhaka', 'Rajshahi', 'Sylhet']
```

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In [21]: # TASK 2: List Comprehension
prices_bdt = [1200, 450, 3200, 890, 5600, 150]

# 2a. Create a new list with only prices above 1000
expensive = [p for p in prices_bdt if p > 1000]
print(f"Expensive items: {expensive}")

# 2b. Create a new list where each price has 15% VAT added
with_vat = [round(p * 1.15, 2) for p in prices_bdt]
print(f"With VAT: {with_vat}")
```

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Expensive items: [1200, 3200, 5600]
```

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With VAT: [1380.0, 517.5, 3680.0, 1023.5, 6440.0, 172.5]
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In [22]: # TASK 3: Tuples & Sets
# 3a. Create a tuple of your (name, age, city) and unpack it
info = ("Kalam", 42, "Cumilla")
name, age, city = info
print(f"{name}, {age}, from {city}")

# 3b. Remove duplicates from this list using a set
raw_ids = [101, 203, 101, 305, 203, 407, 305, 101]
unique_ids = set(raw_ids)
print(f"Unique IDs: {unique_ids}")
```

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Kalam, 42, from Cumilla
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```
Unique IDs: {305, 203, 101, 407}
```

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In [23]: # 4a. Create a dictionary for a student with
#name, department, cgpa, and semester (5 pts)
student = {
    "name": "Kalam",
    "department": "CSE",
    "cgpa": 3.45,
    "semester": 7
}

# 4b. Add a new key "university" with any value (3 pts)
student["university"] = "IIUC"

# 4c. Print all keys and values using a loop (4 pts)
for key, value in student.items():
    print(f"{key}: {value}")
```

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# 4d. Use .get() to print "email" with a default "not provided" (3 pts)
email = student.get("email", "not provided")
print(f"Email: {email}")
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name: Kalam
department: CSE
cgpa: 3.45
semester: 7
university: IIUC
Email: not provided

```
In [24]: # TASK 5: Functions
#5a - average function
def cal_avg(numbers):
    return sum(numbers) / len(numbers)

print(cal_avg([80, 92, 75, 88, 95]))

#5b - grade function
def get_gd(score=0):
    if score >= 90: return "A"
    elif score >= 80: return "B"
    elif score >= 70: return "C"
    elif score >= 60: return "D"
    else: return "F"

print(get_gd(92))
print(get_gd(75))
print(get_gd(45))

#5c - returning multiple values as a tuple
def summarize(scores):
    count = len(scores)
    avg = sum(scores) / count
    return count, avg

count, avg = summarize([85, 90, 78, 92])
print(f"Count: {count}, Average: {avg}")
```

86.0
A
C
F
Count: 4, Average: 86.25

```
In [25]: # 💡 SOLUTION - Task 6
import csv
import json

students = [
    {'name': 'Rahim', 'dept': 'CSE', 'cgpa': 3.65},
    {'name': 'Fatima', 'dept': 'EEE', 'cgpa': 3.82},
    {'name': 'Karim', 'dept': 'CSE', 'cgpa': 3.21},
    {'name': 'Nadia', 'dept': 'BBA', 'cgpa': 3.90},
    {'name': 'Arif', 'dept': 'EEE', 'cgpa': 2.95},
]

# 6a. Write CSV
with open('students.csv', 'w', newline='') as csvfile:
    fieldnames = ['name', 'dept', 'cgpa']
    writer = csv.DictWriter(csvfile, fieldnames=fieldnames)
    writer.writeheader()
    writer.writerows(students)

# 6b. Read CSV
with open('students.csv', 'r') as csvfile:
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reader = csv.DictReader(csvfile)
for row in reader:
    print(f>Name: {row['name']}, Dept: {row['dept']}, CGPA: {row['cgpa']}")

# 6c. Stats → JSON
def calculate_average(numbers):
    return sum(numbers) / len(numbers)

cgpa_list = [s['cgpa'] for s in students]
summary = {
    'total_students': len(students),
    'average_cgpa': round(calculate_average(cgpa_list), 2),
    'highest_cgpa': max(cgpa_list),
    'lowest_cgpa': min(cgpa_list)
}

with open('summary.json', 'w') as f:
    json.dump(summary, f, indent=2)

with open('summary.json') as f:
    result = json.load(f)
print(result)

print('\nHomework Done!')

```

Name: Rahim, Dept: CSE, CGPA: 3.65

Name: Fatima, Dept: EEE, CGPA: 3.82

Name: Karim, Dept: CSE, CGPA: 3.21

Name: Nadia, Dept: BBA, CGPA: 3.9

Name: Arif, Dept: EEE, CGPA: 2.95

{'total_students': 5, 'average_cgpa': 3.51, 'highest_cgpa': 3.9, 'lowest_cgpa': 2.95}

Homework Done!